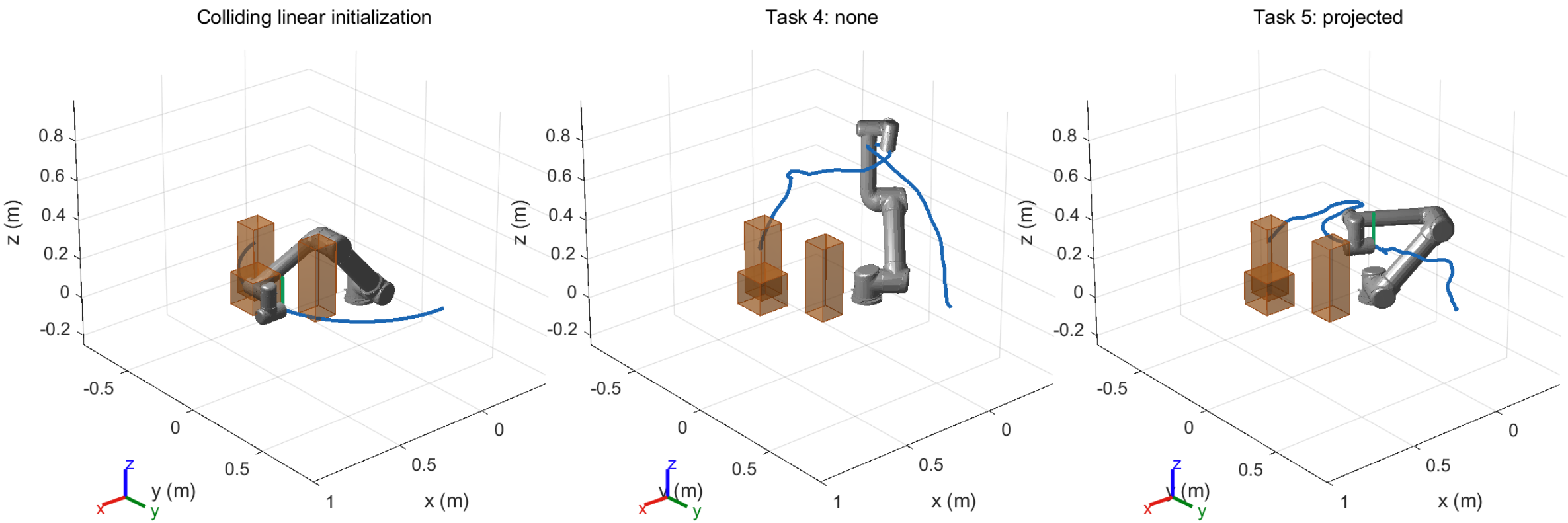
UR5 transfer around three obstacles



Same upright endpoints and colliding linear initialization for both conditions

Development demo: seed 5113; formal results on slide 3
Variant: side pillars move inward 40 mm; right pillar becomes taller

Upright constraint and acceptance

Tool y axis aligned with world +z; free rotation about that axis

$J_{ori} = \lambda \sum_t \|R(q_t)e_y - e_z\|_1 \Delta s$

Selected method: Penalty + PoE Jacobian projection; lambda = 1000

Acceptance: tilt <= 5 deg; conservative clearance >= 35 mm
Fixed endpoints; joint limits; zero world and self collisions

Dense verification: <= 0.0125 rad per joint step; discrete checks

Formal success and uprightness

Scene / condition	Success	Time median (s)	Max tilt median (deg)
main / Task 4	10/10	9.45	108.37
main / Task 5	0/10	149.18	-
variant / Task 4	10/10	8.58	117.24
variant / Task 5	0/10	140.08	-

Common successful pairs: main: n=0; variant: n=0
Upright check: 20/20 constrained runs; penalty-only ablation: 0/10 successful

Seeds 5120-5129; 50 complete + 1 interrupted. Time: complete cases; tilt: successes.

Member C contribution and limits

Added: obstacle scenes, upright mechanism, unified acceptance
Independent replay, paired experiments, figures and animations

Reused: Task 1-3 PoE model, sphere envelopes and STOMP stages

Limits: self collision is enforced only at acceptance
Discrete checks; finite seed sample; no payload or dynamics
Animation progress is a path parameter, not physical time

Reproduce: run_task45 | run_task45_benchmark | verify_task45